Engineering Mathematics - I · Unit_03 · Unit 3

DRAFT — UE26MA141A · Unit_03 · Unit 3 · Modeling of One dimensional Heat equation

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Modeling of One dimensional Heat equation

1 Modeling of One dimensional Heat equation

Study focus: Modeling of One dimensional Heat equation. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000090 · EV0000091

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this session, we will discuss the applications of partial differential equations.

Key details

- And this alpha square is thermal diffusivity.
- It measures the rate at which.
- Heat spreads through a material.
- Observe the diagram.
- The rod is made of one uniform material that is.
- Every part of the rock is made of exact the same substance so that its properties do not change from place to place.
- Let ρ be the density of the material, c be the specific heat of the material and k be the thermal conductivity of the material.
- We have an assumption here.
- That is we assume that the rod's outer surface is perfectly insulated.
- It means heat can only travel along the length of the rod.
- Let u of x comma t be the temperature at position x and at time t.
- The quantity of heat present in the slab.

Example or application from the lecture

- Heat equation is a fundamental partial differential equation in mathematics and physics that describes the distribution of heat in a given region over time.
- Its mathematical formulation is given by dou u by dou t equals alpha square dou square u divided by dou x square.
- Consider a long thin metal rod that has same cross section area that is A along its length.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000090 · EV0000091 · EV0000092 · +1 source units in pack.json

Core explanation · 05:00–09:59

Then the quantity of heat present in the slab is going to be the product of specific heat, how much so it can store, and the mass of the element slab, and the change in temperature.

Key details

- Initially, we assume that C with the specific heat and.
- Change in temperature obviously delta u.
- And this volume is nothing but a times delta x.
- This is what we have seen in the diagram.
- The rate of change of heat in the slab or slice at time t.
- Let R1 be the rate of inflow of heat at X and R2 be the rate of outflow of heat at del X plus delta X.
- So it means.
- We get R C times A rho delta x dou by dou t equals R1 minus R2.
- In one dimension, Fourier's law says that the heat flow, heat flux.
- that is heat per unit area per unit time is directly proportional to the negative of the temperature gradient so we get minus k times dou u by dou x at x so.
- When we multiply the flux by cross-sectional area that is capital A, we are going to get the total rate of heat flow through the phase.
- So and this negative sign is because heat flows from a higher temperature to the lower temperature.
- When you start moving from left to right, it is easy to see that temperature decreases.
- And this is called the thermal diffusivity of the material.
- those value divided by dou x square so where alpha square equals k by 0 is the thermal diffusivity of the material so it talks about heat spreads through a material so on so in this session.

Example or application from the lecture

- Let us discuss R1 and R2.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000094 · EV0000095 · EV0000096 · +1 source units in pack.json

Core explanation · 10:00–14:59

We understood how to derive the expression for dou u by dou t equals dou square alpha square times dou square u by dou x square as a model.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000098 · EV0000099 · EV0000100 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Modeling of One dimensional Heat equation transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/006_Modeling_of_One_dimensional_Heat_equation__32959d21562f5943.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Modeling of One dimensional Heat equation transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/006_Modeling_of_One_dimensional_Heat_equation__32959d21562f5943.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Modeling of One dimensional Heat equation transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/006_Modeling_of_One_dimensional_Heat_equation__32959d21562f5943.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9974; only the JSON segmentation supplies the displayed timestamps.
Modeling of One dimensional Heat equation transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/006_Modeling_of_One_dimensional_Heat_equation__32959d21562f5943.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
2. 1 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
3. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
4. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
5. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
6. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
7. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00030	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0038. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00031	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0038. Sidecar variants compare...
SRC00032	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0038. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00033	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0038. Sidecar variants compare...

C Revision and retrieval practice